

Lemma 3:

iii) G has a cut-vertex a and $G - \{a\}$ has a component H : $\kappa(a) < \kappa(v) \quad \forall v \in V(H)$

$$\Rightarrow \left\{ \begin{array}{l} V(H) = \{b\} \end{array} \right.$$

$$\left\{ \begin{array}{l} \forall (a, c) \in E(G) \quad \kappa(c) < \kappa(b) \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{no } H' \subset G \text{ with } \\ \neq \emptyset \end{array} \right. \left\{ \begin{array}{l} \kappa(a) < \kappa(v) < \kappa(b) \\ \forall v \in V(H') \end{array} \right.$$

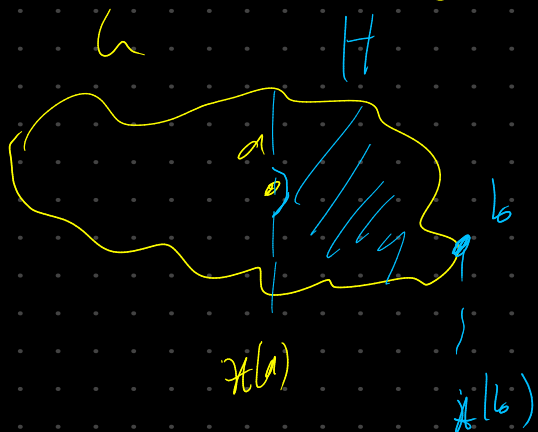
$$\left\{ \begin{array}{l} a \in \overline{N(H')} \neq \{a\} \\ \kappa(v) > \kappa(b) \end{array} \right.$$

$$\forall v \in \overline{N(H')} \setminus \{a\}$$

Proof:

H component of $G - \{a\}$

so that $\kappa(a) < \kappa(v) \quad \forall v \in V(H)$



if $|V(H)| > 1$

apply ii) $H := H - b$

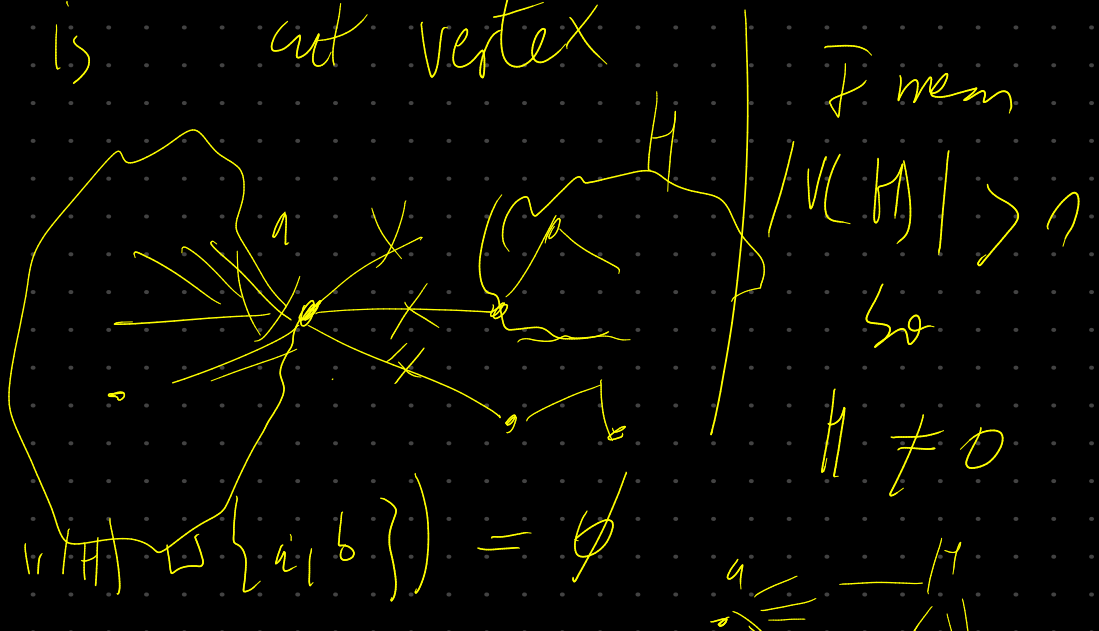
G has no H and $a, b \in V(G) \setminus V(H)$
 such that

$$\begin{cases} x(a) < x(v) < x(b) & \forall v \in V(H) \\ N(H) = \{a, b\} \\ V(G) \setminus (V(H) \cup \{a, b\}) \neq \emptyset \end{cases}$$

if we remove b from our current H
 we will have $\forall v \in V(H) \quad x(a) < x(v) < x(b)$

as $\{a\}$ is cut vertex

$$N(H) = \{a, b\}$$



$|V(H)| > 1$
 so
 $H \neq \emptyset$

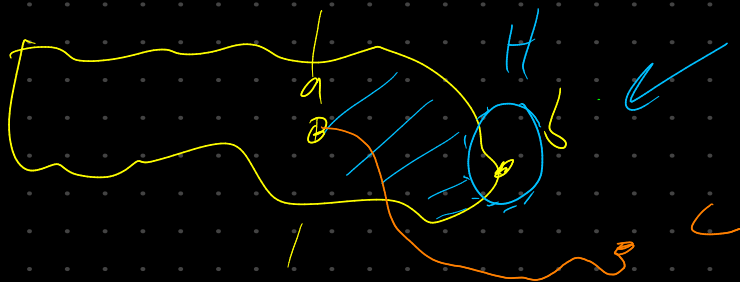


$$V(G) \setminus (V(H) \cup \{a, b\}) = \emptyset$$

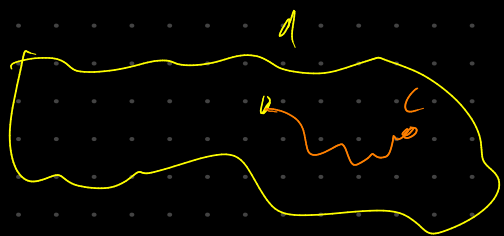
$$V(H) \cup \{a, b\} = V(G)$$

this not possible as x is a cut vertex,
 We can't have one connected component.

Since (iii) requires $H' \neq \emptyset \Rightarrow H \neq \emptyset$
 we are in the case $V(H) = \{b\}$. If
 $(a, c) \in E(H)$ $\chi(b) < \chi(c)$



$N(H) = \{a\}$
 since it got
 split by removing
 vertex b from H .



$H - \{b\}$ is planar
 as it is smaller
 than H .



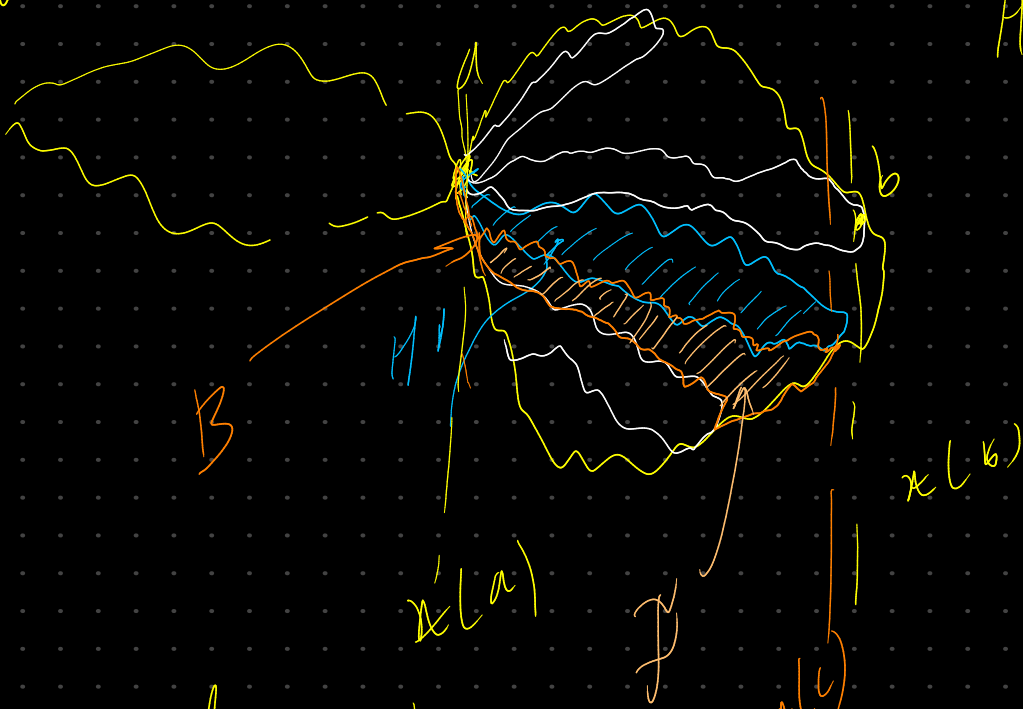
without crossing

> second part of (ii)

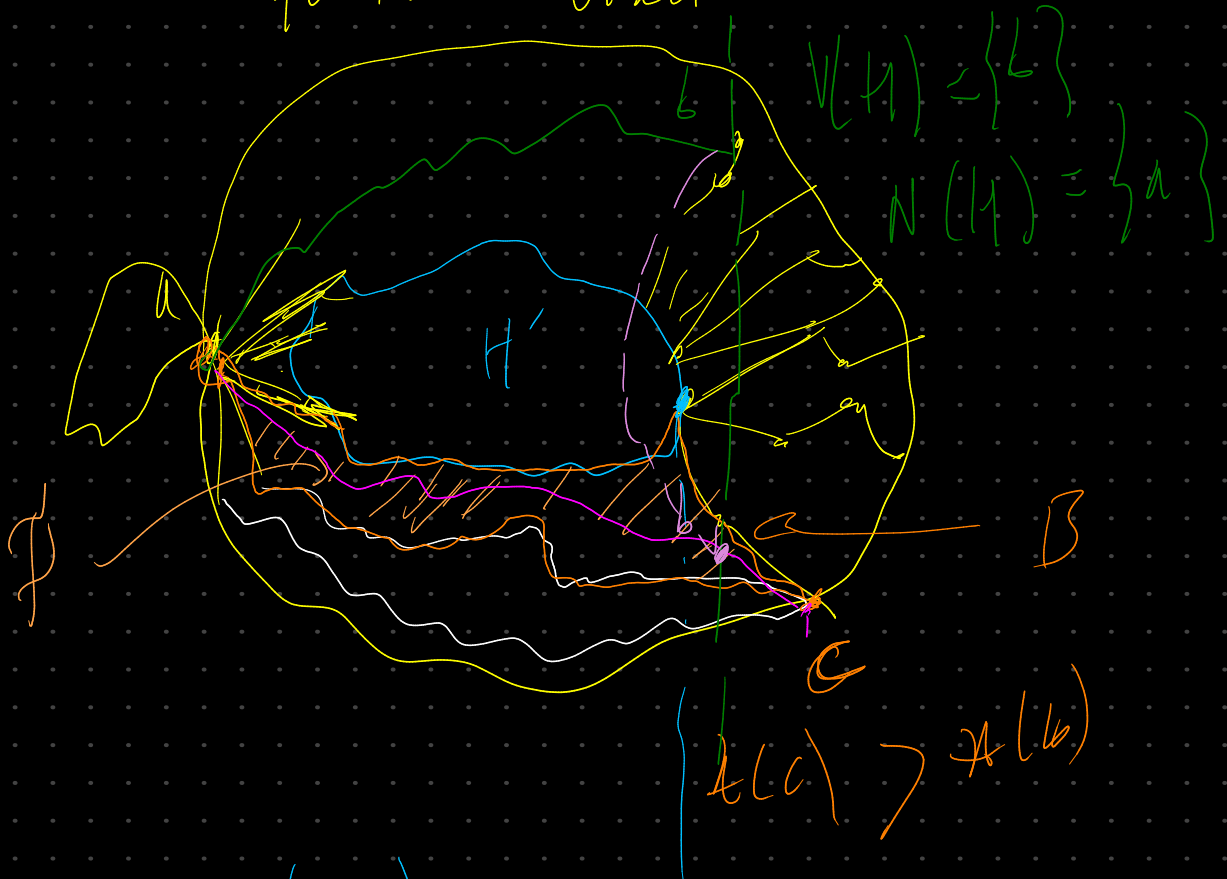
G has no induced subgraph $H' \neq \emptyset$ so that
 $\chi(a) < \chi(v) < \chi(b) \quad \forall v \in V(H)$
 $a \in \{N(H') \neq \{a\}\}'$
 $\chi(v) > \chi(b) \quad \forall v \in N(H') \setminus \{a\}$

by minimality $h - \{b\}$ has an embedding.

H is inside
 h not necessarily
 laying on
 the outside
 face.



B is a closed walk that intersect with H'



$h - \{b\}$ is planar ($\mathcal{C}_h$) so we can use lemma 1
 to introduce $(x(c))$, c will work as $N(H')$ with
 $x(c) > x(b)$ so in the right of $N(H')$

Instead of adding (a, c) we can add (a, b) as a 'sub' edge of (a, c) that we won't draw.

⇒ contradiction as it will result in G being a planar graph.

* The induction phase

What's the use of the minimal counter-example?

Theorem 2:

G has
 π -monotone
with
indep. edges
crossing evenly

→

G has a π -monotone
embedding
with
not changing π -axis
coordinates

min
counter-
example

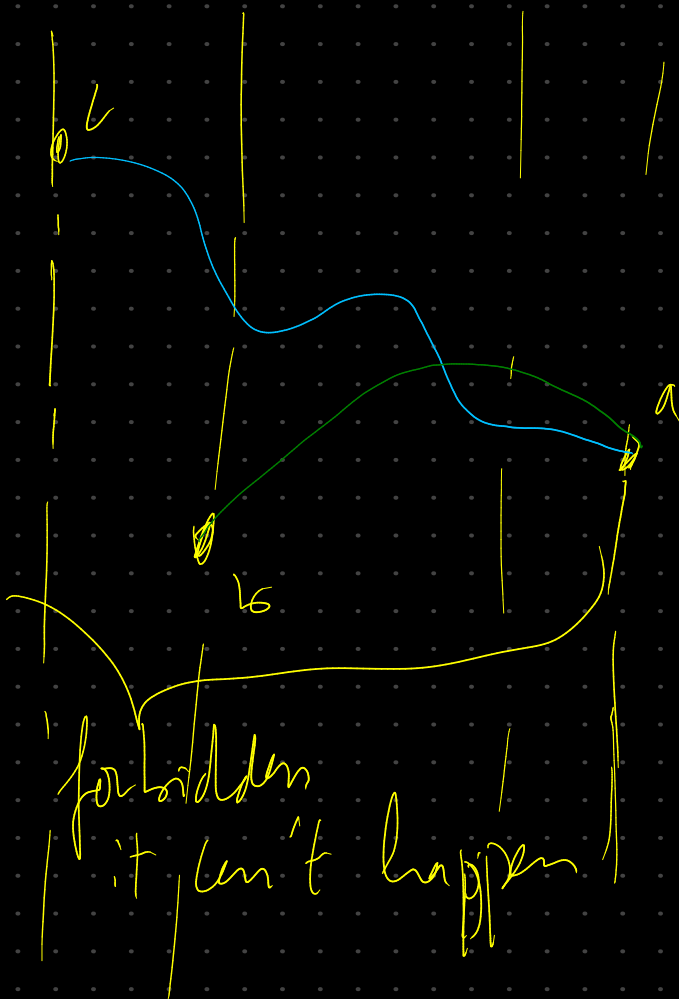
→

(smallest G with
no π -monotone
embedding)

so if we prove that we can't find a minimal counterexample then there is no counterexample so H is true

• we found some properties of minimal counterexample.
if the properties don't fit with the initial state in the theorem it means there is no minimal counterexample.

• if we go back to theorem 2.



we can make it even,
can we not make it even?

for that to be true it needs to be a counterexample to theorem 2 so it needs to have h connectivity

so $H \subset h$ s.t. $a, b \in V(H) \setminus V(H)$
s.t. $\kappa(a) < \kappa(v) < \kappa(b) \forall v \in V(H)$
and $N(H) = \{a, b\}$
and $V(h) \setminus (V(H) \cup \{a, b\}) \neq \emptyset$

the counter-example characteristics help us
 we the case of having odd crossing
 between two edges that share one same
 endpoint

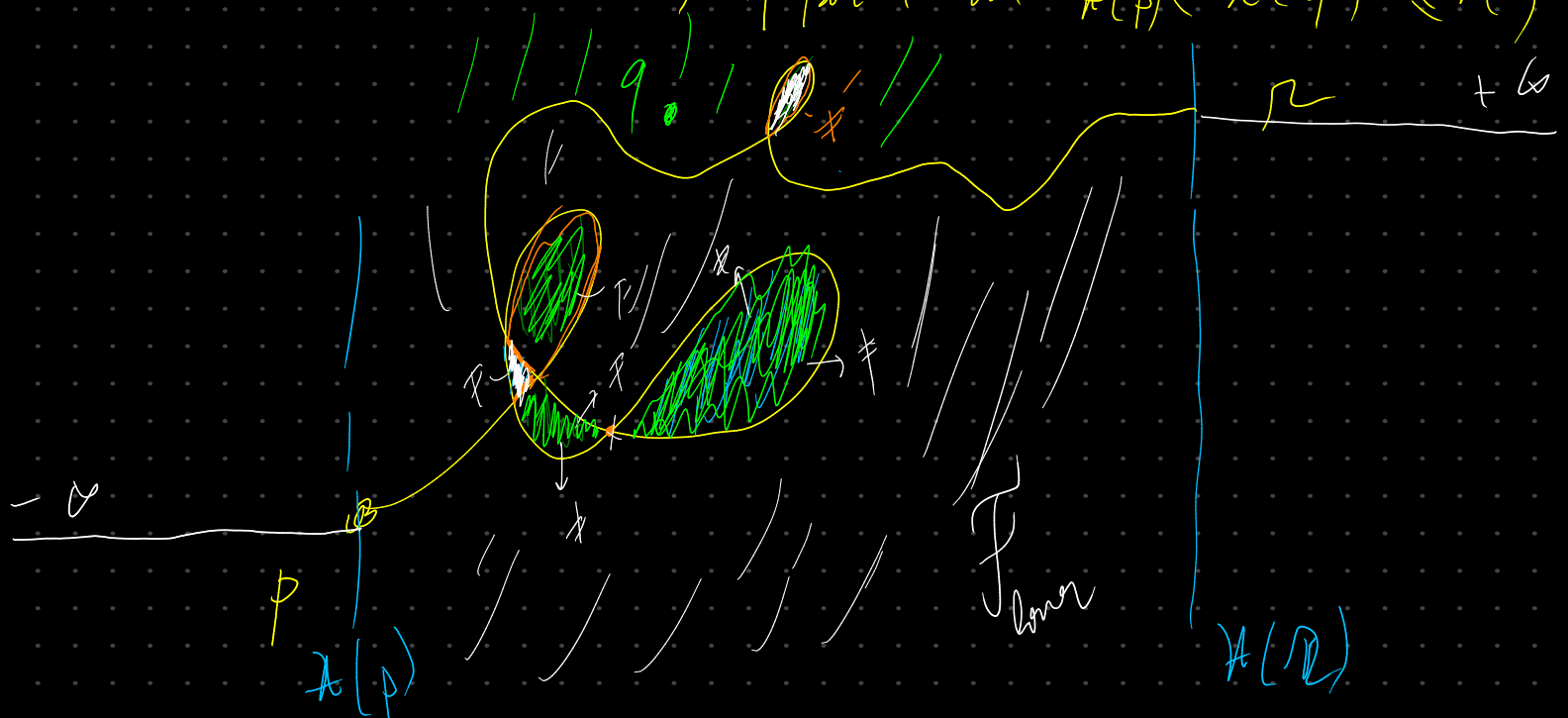
the proof is by induction, and the
 lemma 3 (min-counterexample) help us
 construct the initial phase.

— New terminologies for

lemma 4, 5

laying below/above a curve to curves with self-intersections

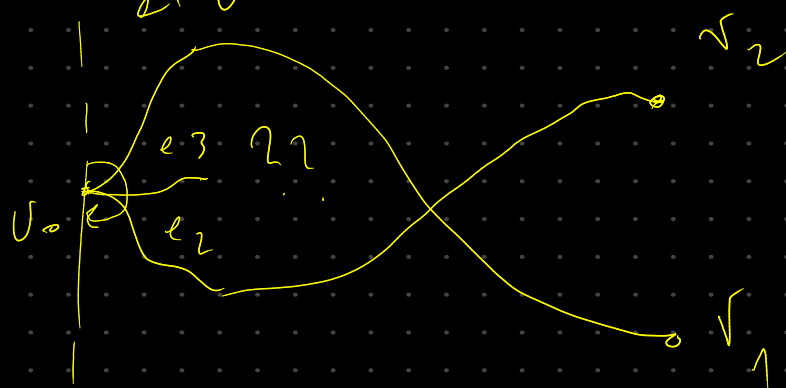
Let C be a curve $\{x \in C \setminus \{p, r\} \mid$
 $q \text{ point with } \#(p) \leq \#(q) \leq \#(r)\}$



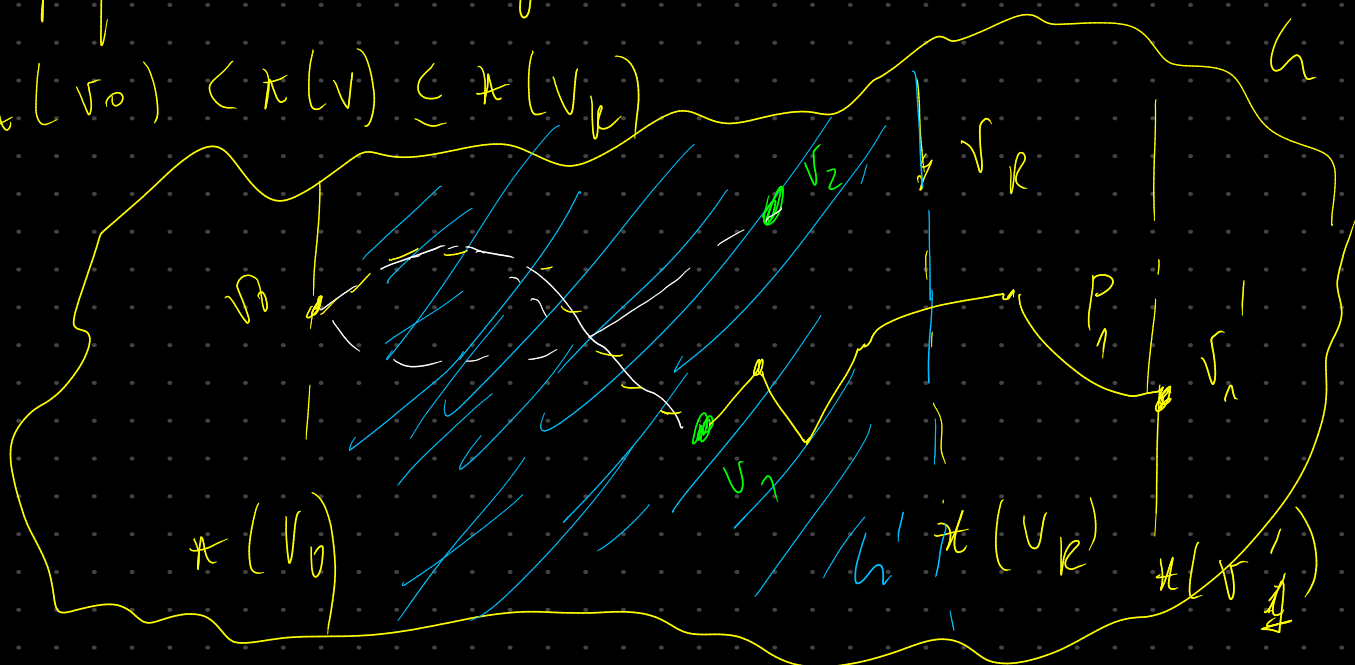
in the following 2 lemmas let h satisfy the assumption of theorem 2. $\forall i \in e, e$

$\begin{matrix} \text{---} e_2 \\ \sim e_3 \\ \text{---} e_1 \end{matrix}$ in the right rot. of v_0 even crossings.

$\begin{matrix} e_1 \sim e_2 \\ e_3 \sim e_2 \end{matrix}$ $\begin{matrix} e_3 \sim e_2 \\ e_3 \sim e_2 \end{matrix}$ evenly



lemma 4: $\kappa(v_n) > \kappa(v_0)$, define h' a graph induced by h on vertices v with $\kappa(v_0) < \kappa(v) \leq \kappa(v_k)$



edges removed in h' .

if h'_1, h'_2, h'_3 are pairwise disjoint and if

$\forall i$ there is a path P_i from v_0 through e_i to some vertex v'_i satisfying $\alpha(v'_i) \geq \alpha(v_2)$

with $\forall v \in P_i$ satisfy $\alpha(v) \geq \alpha(v_0)$

it's bounded left side.

$\Rightarrow$ h'_i has no Neighbors in h to the left of $\alpha(v_0)$



Proof Theorem 2

h be a minimal counterexample to the theorem.

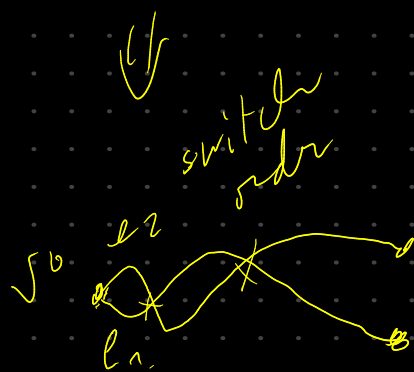
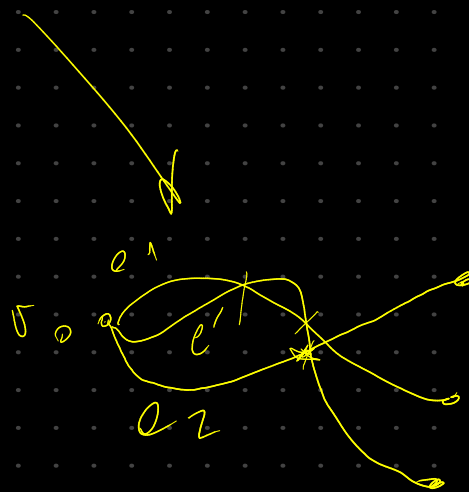
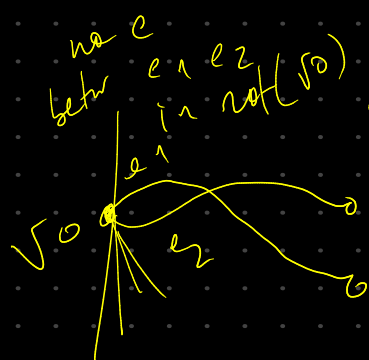
Γ of h minimize the number of odd pairs.

no odd pairs $\Rightarrow$ TH 1 completes the proof

suppose the drawing Γ has odd pairs.

suppose e_1 and e_2 cross oddly.

$\Rightarrow e_1, e_2$ have a shared point

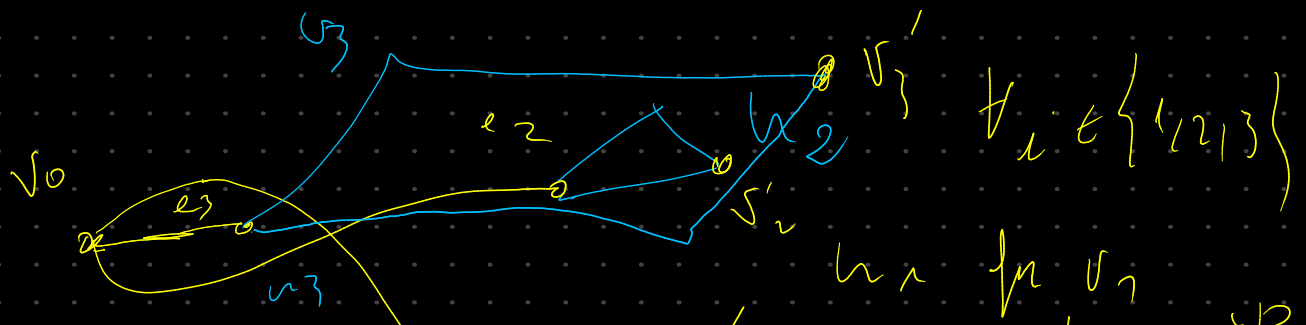


contradict
the choice
of the
drawing of h

suppose

no induced of h
with $\#(v) \geq \#(v_0)$

- Case 1
The right endpoints of
 e_1, e_2, e_3 are in diff.
components of $h_0 - v_0$



$$\begin{cases} |i, j, k| = \{1, 2, 3\} \\ x(v_i) < x(v_j) < x(v_k) \end{cases}$$

v_i is its vertex with largest x -value

Apply lemma (4) $x(v) = x(v_i)$ then h'
induced from h on v with $x(v) < x(v) < x(v_k)$
 $h'_j = \emptyset$, $h'_k = \emptyset$

by lemma 3 (vi) h'_i has only the one vertex
 $v_i = v'_i$ and h'_j and h'_k are non-empty because
 $x(v_i)$ is greater than $x(v_j)$ and $x(v_k)$
 $a = v_0$, $c \in \{v_j, v_k\}$ and $b = v_i$